

Alec Williams

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EDUCATION

University of South Carolina, Columbia, SC

Expected Graduation: May 2027

Bachelor of Science in Computer Science, GPA 3.0

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Computer Networks, Software Engineering, Discrete Mathematics, Advanced Programming in C++, Database Systems

TECHNICAL SKILLS

- Proficient in Java, C++, and Python for developing object-oriented applications and solving complex problems
- Experienced with data structures and algorithms, including implementing linked lists, stacks, queues, and searching/sorting techniques
- Applied socket programming and TCP/IP networking concepts to build client-server applications
- Used Git and GitHub for version control, collaboration, and managing team-based software projects
- Developed user interfaces using JavaFX, FXML, and CSS with a focus on consistent design and usability
- Implemented MVC architecture to organize code and improve maintainability in larger applications
- Performed debugging and testing using tools such as Maven and unit testing frameworks
- Analyzed network traffic and protocols using Wireshark to understand real-world data communication
- Created UML diagrams and applied software engineering principles throughout the development lifecycle
- Demonstrated strong problem-solving skills through algorithm design, optimization, and code testing

EXPERIENCE/PROJECTS

Java Jamboree Coding Platform

GitHub: <https://github.com/Jackdev934/javajambproject>

- Developed a JavaFX-based application with multiple user interfaces and controllers
- Implemented features such as daily question tracking, progress tracking, and bookmarking
- Used Git and GitHub for version control and team collaboration
- Designed UI using FXML and CSS for consistent layout and styling
- Debugged and tested application features using Maven and unit testing

Terminal Messaging Application

- Developed a terminal-based messaging system in Java using socket programming
- Implemented client-server communication for sending and receiving messages in real time
- Applied networking concepts such as TCP connections, ports, and data transmission
- Debugged connection issues and ensured reliable message delivery between users
- Tested functionality through multiple client interactions and edge cases